① $P(A) = 0.6 \quad P(B) = 0.5$ $\qquad P(A \cup B)^c = 0.30$

ⓐ $P(A \cup B) = 0.7$

ⓑ $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.4$

ⓒ $P(A \cap B^c) = P(A) - P(A \cap B) = 0.2$

ⓓ $P(A - B) = P(A \cup B) - P(A \cap B) = 0.3$

② $S = \{(a,b) : 1 \le a, b \le 6, \ a, b \in \mathbb{N}\}$

$A \to x_1 + x_2 = 8$

2 6
8 5
4 4
5 3
6 2

$\to 5/36$

③ A → ~~atleast a girl~~ randomly chosen is girl

B → all girl

$P(B|A) \quad \dfrac{P(B \cap A)}{P(A)} \quad \dfrac{\frac{1}{2^n}}{1 - \frac{1}{2^n}\frac{1}{2}} = \dfrac{1}{2^n - 1} \quad \dfrac{1}{2^{n-1}}$

(5) $Cov(Z,W) = E(ZW) - E(Z)E(W)$.

$ZW = (1+x)(1+x+xy^2)$

$= 1+x+xy^2+x+x^2+x^2y^2$

$= 1+2x+x^2+xy^2+x^2y^2$

$E(ZW) = 1 + E(x^2) + E(x^2)E(y^2)$

$= 3$

$E(Z) = 1 \quad E(W) = 1$

$cov(Z,W) = 3-1 = 2$.

(6) $A_i \rightarrow$ event of receiving offer letter from the i^{th} company.

$\mathbb{P}\left(\bigcup_{i=1}^{4} A_i\right) = 1 - \mathbb{P}\left(\cap A_i^c\right)$

$= 1 - \prod_{i=1}^{4} \mathbb{P}(A_i^c)$

$= 1 - (0.8)^4$

$= 0.59$

(7) $I_i \rightarrow$ indicator variable

$$I_i = \begin{cases} 1, & p=0.1 \\ 0 & p=0.9 \end{cases}$$

$$X = \sum_{i=1}^{1000} I_i$$

$P(X \geq 120) = 1 - P(X < 120)$

$$P(X < 120) = \sum_{i=1}^{119} {}^{1000}C_i (0.1)^i (0.9)^{1000-i}$$

(or)

$$P\left(\frac{X - n\mu}{\sqrt{n} \times \sigma} \geq \frac{120 - n\mu}{\sqrt{n}\,\sigma}\right)$$

$$P\left(N(0,1) \geq \frac{120 - 1000 \times 0.1}{\sqrt{1000}\sqrt{(0.1 - 0.1^2)}}\right)$$

$P(N(0,1) \geq 2.108) = 0.0175$

(8) $X_i \to$ no. of sandwiches by each guest.

$$X_i = \begin{cases} 0 & , 1/4 \\ 1 & , 1/2 \\ 2 & , 1/4 \end{cases}$$

$\mu = \frac{1}{2} + \frac{1}{2} = 1$

$\sigma^2 = \frac{1}{2} + 1 - 1 = \frac{1}{2}$

$\sigma = \frac{1}{\sqrt{2}}$

$P(\Sigma X_i > k) < 5\%$

$$P\left(\frac{\Sigma X_i - n\mu}{\sqrt{n}\cdot\sigma} > \frac{k - n\mu}{\sqrt{n}\,\sigma}\right) < 5\%$$

$$P\left(N(0,1) > \frac{k - 64 \times}{8 \times \frac{1}{\sqrt{2}}}\right) < 5\%$$

$$\frac{k - 64}{\frac{8}{\sqrt{2}}} = 1.645 \quad , \quad k = 1.645 \times \frac{8}{\sqrt{2}} + 64 = 73.30 \approx 74$$

(9) The marginals are also gaussian

$x \sim N(0,1) \qquad y \sim N(0,1)$.

$E(X) = E[Y] = 0$

$var(x) = var(y) = 1$

$cov(xy) = \rho$

(10) $f_{x|y} = \dfrac{f_{xy}}{f_y} \qquad \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$

$x \sim N(1, 4)$

$y \sim N(2, 3)$

$$f_{xy} = \frac{1}{2\pi \times \sqrt{11}} e^{-\frac{(3x^2 + 4y^2 - 2xy - 2x - 14y + 15)}{11 \times 2}}$$

$C \to \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \quad |C| = 11 \qquad \text{cofa} \begin{pmatrix} 3 & -1 \\ -1 & 4 \end{pmatrix}$

$C^{-1} \; \frac{1}{11} \begin{pmatrix} 3 & -1 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} x-1 \\ y-2 \end{pmatrix}$

$(x-1, y-2)$

$$\frac{1}{11} \begin{bmatrix} 3x-3-y+2 & -x+1+4y-8 \end{bmatrix} \begin{pmatrix} x-1 \\ y-2 \end{pmatrix}$$

$3x-y-1 \qquad -x+4y-7$

$$\frac{1}{11} \left((3x-y-1)(x-1) + (y-2)(-x+4y-7) \right)$$

$3x^2 \cancel{-3x} - 3x - xy + y \cancel{-x} + 1 - xy + 4y^2 - 7y + \cancel{2x} - 8y + 14$

$$f_y = \frac{1}{\sqrt{2\pi \times 3}} e^{-\frac{(y-2)^2}{6}}$$

$$\frac{f_{xy}}{f_y} = \frac{1}{\sqrt{2\pi} \times \sqrt{\frac{11}{3}}} e^{\frac{-3x^2-4y^2+2xy+2x+14y-15}{22} + \frac{y^2+4-4y}{6}}$$

$$e^{\frac{-7x^2-24y^2+12xy-32x+84y-46}{66}}$$

$$\frac{-(7x^2+x(32-12y)+24y^2-84y+46)}{66}$$

$9x^2 + 2x(6-6y)$ +

$$\frac{2x^2-12y^2+6xy-38x+42y-1}{66}$$

$$\frac{-9x^2-12y^2+6xy+6x+42y-45+11y^2+44-44y}{66}$$

$-9x^2-y^2+6xy+6x-2y-1$

$$\frac{-(9x^2-6xy-6x+y^2+2y+1)}{66}$$

$$\frac{9x^2-2\cdot 3x(y+1)+(y+1)^2}{66} \Rightarrow \frac{(3x-(y+1))^2}{66}$$

$\Rightarrow \boxed{\text{mean} = \frac{y+1}{3} \quad \text{var} = \frac{11}{3}}$ $\Rightarrow \frac{\left(x - \frac{(y+1)}{3}\right)}{66/9}$

$f_{x|y} \sim \mathcal{N}\left(\frac{y+1}{3}, \frac{11}{3}\right)$

(ii) $z = [3 \ \ -2] \begin{bmatrix} x \\ y \end{bmatrix}$

$\mu' = A\mu \qquad C' = ACA^T.$

$[3 \ \ -2] \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ | $C' = [3 \ \ -2] \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$\mu' = 0$

$C' = [1 \ \ -5] \begin{bmatrix} 3 \\ -2 \end{bmatrix} = 13$

$\Rightarrow \quad f_z = \frac{1}{\sqrt{2\pi \times 13}} e^{-\frac{a^2}{26}}.$

$$\boxed{f_z = \frac{1}{\sqrt{26\pi}} e^{-a^2/26}}$$

$\text{Corr}(z, x) = \frac{\text{cov}(z, x)}{\sigma_x \sigma_z} = \boxed{\frac{1}{\sqrt{13}}}$

$\frac{E[zx]}{\sqrt{13}} = \frac{E[3x^2 - 2xy]}{\sqrt{13}} = \frac{3-2}{\sqrt{13}} = \frac{1}{\sqrt{13}}$

(12) The conditional is also gaussian and has mean $\mu_{(1)} + \Sigma_{12}\Sigma_{22}^{-1}(x_{(2)} - \mu_{(2)})$

variance $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$

here $\rightarrow$ $\mu_{(1)} = 0$ $\quad \mu_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Sigma_{11} = (4) \quad \Sigma_{22} = \begin{pmatrix} 3 & -1 \\ -1 & 5 \end{pmatrix}$

$\Sigma_{12} = (2, 1) \quad \Sigma_{21} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\Rightarrow \mu_{1|2} = (2,1) \times \frac{1}{14}\begin{pmatrix} 5 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} y \\ z \end{pmatrix}$

$\frac{1}{14}\begin{pmatrix} 11 & 5 \end{pmatrix}\begin{pmatrix} y \\ z \end{pmatrix}$

$$\boxed{\mu_{1|2} = \frac{1}{14}(11y + 5z)}$$

$\mathrm{var}_{1|2} = 4 - (2,1)\frac{1}{14}\begin{pmatrix} 5 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$4 - \frac{1}{14}\begin{pmatrix} 11 & 5 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$4 - \underline{27} = \boxed{2.07}$